Date of publication xxxx 00, 2026, date of current version xxxx 00, 2026.

Digital Object Identifier 10.1109/ACCESS.2017.DOI

An AI-Ready Air-to-Ground Channel Model for HAPS–UAV Access Networks

HARSHA MATADHIKARI[1]
[1]PG Diploma in Advanced Communication Engineering with Quantum AI Integration, Indian Institute of Technology Delhi, New Delhi, India (e-mail: harsha.matadhikari@gmail.com)

Corresponding author: Harsha Matadhikari (e-mail: harsha.matadhikari@gmail.com).

This work was carried out as part of the PG Diploma in Advanced Communication Engineering with Quantum AI Integration, Indian Institute of Technology Delhi.

ABSTRACT This paper builds an AI-ready channel model for a HAPS–UAV–Ground access network and uses it to train a learned user–platform association policy. The model combines free-space and ITU-R atmospheric loss for the 38 GHz backhaul and feeder links, building-blockage path loss for the 2 GHz service and access links, and multi-HAPS co-channel interference under a 3-HAPS constellation, showing that angular position affects SINR more than distance does. An LSTM forecasts feeder-link rain attenuation six hours ahead; once properly early-stopped (an earlier, longer-trained version was overfit), it beats a persistence baseline at every horizon except the shortest, where a validated weighted blend of the two adds a further small improvement. A Deep Q-Network trained on this forecast, SINR, distance and battery state is compared against max-SINR and max-throughput rules on held-out weather, with the UAV relay modeled realistically as a single static platform rather than one that tracks each user. The learned policy matches both rule-based baselines on reward while uniquely retaining and using the relay path as a targeted fallback for users near outage, a capability the baselines cannot replicate.

INDEX TERMS HAPS, UAV, non-terrestrial networks, air-to-ground channel, LoS probability, SINR, multi-node interference, UAV relay, backhaul link, rain attenuation, LSTM forecasting, deep reinforcement learning

I. INTRODUCTION

A High-Altitude Platform Station (HAPS) is a stratospheric, solar-powered platform at approximately 20 km altitude with a fixed service footprint of up to 100 km radius [1], [2]. It connects to the ground over three link types:

- **Service link** (HAPS to UE, 2–2.2 GHz [30]): direct, long-range connectivity without specialized ground hardware.
- **Backhaul link** (HAPS to Gateway, 38.0–39.5 GHz Ka-band, WRC-19 [29]): high-capacity ground-gateway connectivity.
- **Relay link** (HAPS to UAV to UE): a two-hop backup path for users with a poor direct link – a 38 GHz feeder hop (HAPS–UAV, same band as the backhaul) followed by a 2 GHz access hop (UAV–UE) from a UAV hovering at lower altitude.

The 38 GHz backhaul/feeder links incur significant atmospheric loss (rain, fog, gas), while sitting in a spectral valley between the 22.235 GHz water-vapor and 60 GHz oxygen absorption resonances [6]. The 2 GHz service/relay links are resilient to atmospheric effects [4] but sensitive to building blockage. Because the 100 km service radius pushes many links to low elevation angles, slant-path lengths and the resulting atmospheric loss grow accordingly [4].

Distance-only link budgeting, however, misses a second, larger effect: with three HAPS sharing spectrum, co-channel interference from the other two platforms makes SINR depend strongly on *angular position*, not just distance. A static analysis cannot react to this, or to time-varying rain fades on the 38 GHz backhaul. This motivates an AI-ready model: a validated channel model, an LSTM rain forecaster, and a DQN agent that learns real-time user–platform association.

Research flow. The remainder of this paper follows three stages: (1) a *channel and attenuation model* – free-space path loss, ITU-R rain/fog/gas/scintillation, and

multi-HAPS interference (§III–§V), characterised numerically in §VI; (2) *baseline association policies* – random, max-SINR, and max-throughput rules, evaluated on that channel model to establish how much association choice alone matters (§VI-E); and (3) a *learned association policy* – an LSTM rain forecaster (§VII) feeding a DQN agent (§VIII), evaluated against the stage-2 baselines (§IX).

The results follow the same three stages: (1) a reproducible ITU-R-based channel model for the HAPS, UAV and gateway links, including a quantitative characterisation of multi-HAPS interference showing where it limits the 2 GHz service link and why the 38 GHz backhaul does not, and the HAPS→UAV→UE relay path as an alternative to the direct service link; (2) three rule-based association baselines (random, max-SINR, max-throughput) evaluated on that model; and (3) an LSTM rain forecaster benchmarked out-of-sample against persistence, feeding a DQN association agent trained on 2024 weather and evaluated on held-out 2025 days against the stage-2 baselines.

II. RELATED WORK

Building-blockage/LoS modeling (2 GHz access tier). Holis and Pechac [18] give the elevation-dependent building-blockage shadowing model this paper adopts for the 2 GHz service link and the relay's access hop, originally formulated for a single serving platform with no co-channel interference term; ITU-R P.1410 [11] offers a simpler elevation-angle-only LoS baseline that the Holis–Pechac model refines with building geometry. The 38 GHz backhaul/feeder link does not need this model: its near-nadir geometry (§III) avoids building obstruction, so atmospheric loss (rain/fog/gas, §IV) is the dominant impairment there instead.

Multi-HAPS co-channel interference. Arani *et al.* [17] study HAPS–UAV heterogeneous networks with a DRL controller, closest in spirit to the association problem here, but evaluate a single HAPS–UAV pair with a time-invariant channel and a throughput-only objective; we make this delta explicit on three points. (i) *Interference:* their controller has no co-channel term to learn around, while our 3-HAPS geometry shows angular position alone swings SINR by 9.94 dB at fixed range (§V), a degree of freedom their setting cannot expose. (ii) *Weather coupling:* their feeder/access channel is time-invariant, so the controller cannot anticipate fades; our DQN consumes a six-horizon LSTM rain forecast (§VII) specifically because the feeder link is not static. (iii) *Reward structure:* we jointly penalize outage, fairness, and per-platform battery depletion (§VIII, Eq. (9)) in a single reward, rather than optimizing throughput alone, so the comparison against rule-based max-throughput/max-SINR baselines (§IX) isolates what the extra state (forecast, battery, fairness) buys over a narrower throughput-only objective. Liu *et al.* [33] optimize sum-rate and worst-case SINR across multiple cooperating HAPS and ground base stations via joint user association and beamforming, but likewise treat the channel as static and do not model time-varying rain fades or an onboard forecaster; our channel model instead quantifies the *angular* SINR heterogeneity this cooperative setting would need to counteract, and complements it with the learned, weather-aware policy their static optimization does not provide.

Learned resource allocation and association in NTN. Naous *et al.* [32] survey reinforcement learning across non-terrestrial platforms (satellites, HAPS, UAVs) and note that most RL-based association work either assumes a static channel or omits weather-driven feeder-link variability – the gap this paper's LSTM-fed DQN targets. Foundational DQN [25] and RL theory [26] underpin the agent design of §VIII; PPO [27] was considered but DQN suits this paper's small discrete (six-action) space better, at lower sample complexity. Fairness is quantified with the Jain index [28], and per-link SINR is combined in the linear domain following standard practice [22] rather than by averaging in dB.

Rain attenuation forecasting. Cornejo *et al.* [34] train an LSTM on synthetic, ITU-R-derived rain-attenuation traces for satellite links, reporting improved short-term prediction accuracy over classical models; this paper instead trains on several years of real, multi-source weather data (NASA POWER, METAR, ERA5, 2023–2026) for a specific HAPS feeder-link geometry and benchmarks against a persistence baseline, which (§VII) proves non-trivial to beat at short horizons. The resulting rain margins are consistent with the Ka-band FSS rain-margin guidance in ECC Report 156 [20] (§IV-A).

Survey context. The recent survey by Svistunov *et al.* [31] catalogs HAPS channel modeling, interference control, and AI-driven resource allocation at a high level, and the broader survey by Bakambekova *et al.* [19] on AI for space-air-ground integrated networks identifies weather-aware, cross-layer learning as an open direction at the SAGIN level; this paper provides one of the quantitative instances – a validated channel model and a trained, evaluated association policy – that both surveys point to as needed.

Positioning. None of the above combines all three elements this paper does: (i) an ITU-R-validated, multi-HAPS interference-aware channel model showing that angular position can dominate distance in SINR, (ii) an out-of-sample rain forecaster feeding (iii) a DQN association policy evaluated against rule-based baselines on real held-out weather. This combination, rather than any single component in isolation, is the contribution.

III. SYSTEM TOPOLOGY AND CHANNEL MODEL

Three HAPS nodes and one UAV relay serve ground users (access layer); a single ground gateway aggregates all three backhaul links (backhaul layer) into the core network (Table 1, Fig. 1). Elevation angle and slant-

TABLE 1: Simulated network configuration (AGL = above ground level)

Node	Altitude	Count	Spacing
HAPS	20 km	3	Equilateral 65 km, 1 server, 2 interferers
UAV relay	500 m AGL	1	Static, fixed at HAPS-triangle centroid (no mobility)
Gateway	50 m AGL	1	Centroid of HAPS triangle
Users	1.5 m (hand-held)	N (configurable)	Uniform random in coverage

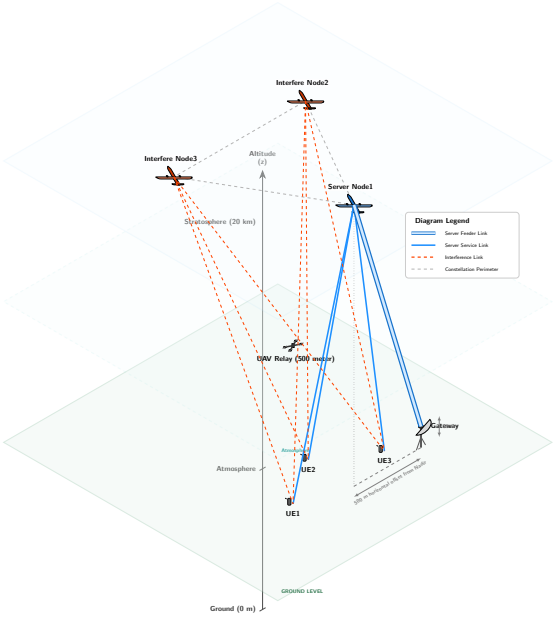


FIGURE 1: Network topology: three HAPS nodes and one UAV relay serve ground users over an access layer; a single ground gateway aggregates the backhaul links into the core network.

path length follow directly from each link's altitude pair and horizontal offset: the backhaul link (HAPS to a near-nadir gateway) and the HAPS–UAV feeder hop are both near-zenith (≈88.5–89.9° at nadir); the UAV–UE access hop is dominated by horizontal offset and building blockage rather than altitude; and the direct HAPS–UE service link spans the full service radius, from near-zenith at nadir to low elevation at the 100 km edge.

Channel model. Three propagation models apply to this topology: (i) free-space path loss (FSPL), common to all links; (ii) ITU-R rain, fog, gas and scintillation losses, applicable to the 38 GHz backhaul and feeder links; and (iii) a building-blockage LoS/NLoS probability model (Holis & Pechac [18]), applicable to the 2 GHz UAV/HAPS access-tier links.

The ITU-R P.530 free-space path loss [3] is

$$L_{\mathrm{FSPL}} = 32.44 + 20\log_{10} f_{\mathrm{MHz}} + 20\log_{10} d_{\mathrm{km}}\ [\mathrm{dB}], \quad (1)$$

where f_{MHz} is the carrier frequency and d_{km} the link distance. This is algebraically the wavelength form $L_{\mathrm{FSPL}}(h,r) = 20\log_{10}(4\pi d_{h,r}/\lambda)$ used later in the multi-HAPS SINR formulation (§V): substituting $\lambda = c/f$ and converting units folds the resulting constant into the 32.44 dB term above.

IV. ATMOSPHERIC AND PROPAGATION MODELS

A. 38 GHZ BACKHAUL/FEEDER LINK

Rain attenuation γ_R (dB/km) follows the ITU-R P.838 power law [5]

$$\gamma_R = k(f,\theta)\, R^{\alpha(f,\theta)} \quad [\mathrm{dB/km}], \quad (2)$$

with $k \approx 0.392$, $\alpha \approx 0.869$ at 38 GHz for the near-nadir (≈ 89.86°) HAPS–gateway geometry (horizontal polarisation), giving $\gamma_R \approx 2.9$ dB/km at 10 mm/h and ≈ 6.4 dB/km at 25 mm/h. Multiplying by the ITU-R P.618 effective slant length L_{eff} [4] gives $A_{\mathrm{rain}} = \gamma_R L_{\mathrm{eff}}$. For Delhi's $R_{0.1} = 15.7$ mm/h rain rate (ITU-R P.837 [10]), this gives ≈ 19 dB of rain loss on the near-nadir gateway path. Against the clear-sky backhaul SINR of ≈ +38 dB (§V) and ≈ 0.4 dB of additional gas loss, the resulting link margin is $38 - 19 - 0.4 \approx 18.6$ dB at 99.9 % availability – consistent with the 15–25 dB Ka-band rain margins reported for FSS links by ECC Report 156 [20] – so the 38 GHz backhaul therefore closes comfortably at the design availability target.

Fog/cloud loss (ITU-R P.840 [7]) and gas absorption (ITU-R P.676 [6]) are modeled as specific attenuations times slant length, $A_{\mathrm{cloud}} = \gamma_c(f,M,T)L_{\mathrm{slant}}$ and $A_{\mathrm{gas}} = (\gamma_o + \gamma_w)L_{\mathrm{slant}}$; both are small at 38 GHz (fog typically < 0.2 dB; clear-sky gas ≈ 0.2–0.3 dB/km, versus ∼15–20 dB/km at the 60 GHz oxygen resonance). Tropospheric scintillation (ITU-R P.618 Annex 1 [4]) is

$$A_{\mathrm{scint}}(p) = a(p)\cdot\sigma_{\mathrm{pre}}(f,\theta,\mathrm{climate}) \quad [\mathrm{dB}], \quad (3)$$

growing with frequency and shrinking with elevation. Total backhaul/feeder slant-path attenuation sums all terms:

$$L_{\mathrm{total}} = L_{\mathrm{FSPL}} + A_{\mathrm{rain}} + A_{\mathrm{cloud/fog}} + A_{\mathrm{gas}} + A_{\mathrm{scint}}. \quad (4)$$

B. 2 GHZ SERVICE/RELAY LINK

Unlike the 38 GHz links, the 2 GHz band is resilient to atmospheric effects but sensitive to building blockage. For a UAV or HAPS node u at horizontal distance $r_{u,k}$ from user k, the LoS probability follows the building-blockage product series of Arani et al. [17], parameterised by Holis & Pechac [18]:

$$P_{u,k}^{\mathrm{LoS}} = \prod_{n=0}^{J}\left[1 - \exp\left(-\frac{1}{2\xi^2}\left(h_u - \tfrac{n+\frac{1}{2}}{J+1}(h_u - h_k)\right)^2\right)\right], \quad (5)$$

with $J = \lfloor \frac{r_{u,k}}{1000}\sqrt{\alpha\beta} - 1 \rfloor$ building layers, and environment-dependent parameters (α,β,ξ) (e.g. sub-urban $0.1, 750, 8$ m; urban $0.3, 500, 15$ m; dense urban

0.5, 300, 20 m; high-rise 0.5, 300, 50 m) [18]. This single-relay topology (Table 1) needs only a 3-HAPS interference sum; no UAV-to-UAV term is required.

Each propagation state $z \in \{\text{LoS}, \text{NLoS}\}$ adds a log-normal shadowing term, averaged in linear gain (not dB, since the two states can differ by up to ∼17 dB in dB-domain averaging):

$$\overline{g} = 10^{-\text{FSPL}/10}\left[P_{\text{LoS}}\, e^{\frac{(a\sigma_{\text{LoS}})^2}{2}} + P_{\text{NLoS}}\, e^{-a\mu(\theta)+\frac{(a\sigma(\theta))^2}{2}}\right],$$
$$\overline{\text{PL}} = -10\log_{10}\overline{g}, \qquad a = \tfrac{\ln 10}{10}. \tag{6}$$

The downlink SINR at user k from node u is then

$$\gamma_{u,k} = \frac{p_u\, g_{u,k}}{\sum_{u' \neq u} p_{u'} g_{u',k} + \sigma_0^2}, \qquad g_{u,k} = 10^{-\overline{\text{PL}}/10}. \tag{7}$$

V. MULTI-HAPS INTERFERENCE, SINR, AND THE RELAY PATH

With three HAPS sharing spectrum, a serving HAPS h provides the desired signal while the other two contribute co-channel interference at the same receiver r (gateway, UAV, or UE):

$$\gamma_{h,r} = \frac{P_h G_h G_r g_{h,r}}{N_r + \sum_{h' \neq h} P_{h'} G_{h'} G_r g_{h',r}}, \tag{8}$$

with channel gain $g_{h,r} = 10^{-L^{\text{tot}}_{h,r}/10}$ from the total propagation loss of §IV. This same formulation instantiates the backhaul link (r = gateway), the feeder hop (r = UAV), and the direct service link (r = UE); the UAV–UE access hop uses UAV transmitters/receivers in place of HAPS and is interference-free with a single relay.

Backhaul link. The gateway sits near the serving HAPS's nadir, while the two interfering HAPS are ≈65 km away and seen off-boresight through sidelobes (ITU-R F.699, floored at −10 dBi). With directional antenna gains included ($G_h = 13.8$ dBi, $G_g = 39.7$ dBi [14]), the interference term is ≈ −35 dB below the noise floor, so the backhaul SINR is ≈ +38 dB and **noise-limited**, essentially flat across the 10–500 m gateway-offset range considered.

Feeder hop (HAPS→UAV). The single relay UAV is fixed at the HAPS triangle's centroid (§V-A), not tracking the user. This hop therefore has one **constant** geometry, regardless of distance or which HAPS relays: ≈27.5° elevation, noise-limited at +30.7 dB. Antenna gains are $G_h = 13.8$ dBi and a UAV-mountable mmWave patch array $G_u = 30.7$ dBi [15].

Service link (HAPS→UE). Here the picture is different. A receiver at horizontal distance r from the serving HAPS's nadir and the two interferers at ≈65 km produces an **angularly-dependent** SINR: at a fixed $r = 50$ km, SINR ranges from −7.40 dB toward an interferer ($\theta = 0°$) to +2.54 dB away from the constellation ($\theta = 210°$) – a 9.94 dB spread – mapping to a good sector (≈120°–290°) and a bad sector (≈330°–75°) (Fig. 6, §VI). Averaged over all azimuths, mean service-link SINR falls from +6.3 dB at 10 km to −4.3 dB at 100 km, while the angular min–max band widens from 1.95 dB to a peak of 11.94 dB at 75 km (Fig. 7, §VI) – beyond ∼30 km, angular position dominates over distance decay. This spread (9.94 dB at fixed distance) exceeds the ∼2–3 dB cost of simply doubling the link distance, so a distance-only link budget cannot characterise the interference environment, and the network needs TDMA/FDMA/beamforming/power control, or a learned association policy, to manage it.

A. RELAY PATH ANALYSIS (HAPS→UAV→UE)

The relay path's end-to-end capacity is the decode-and-forward bottleneck of its two hops, $C_{\text{relay}} = \min(C_{\text{haps-uav}}, C_{\text{uav-ue}})$. The single relay UAV (Table 1) is static, fixed at the HAPS triangle's centroid rather than hovering beside whichever user it serves – the only placement consistent with "static" and with there being one UAV to assign (§VIII). For an equilateral 65 km constellation this centroid is equidistant from all three HAPS (circumradius $65/\sqrt{3} \approx 37.5$ km), so the feeder hop (HAPS→UAV) has one **constant** elevation (≈27.5°) and capacity (10.2 bits/s/Hz) regardless of which HAPS relays or which user is served. The access hop (UAV→UE, 2 GHz) instead depends on each user's real horizontal distance from this one fixed point: capacity falls from 11.1 bits/s/Hz at the UAV's own location to 0.96 bits/s/Hz at 5 km and 0.17 bits/s/Hz at 10 km, collapsing far faster than the direct HAPS→UE link does over the same distances because the UAV's 500 m altitude loses elevation – and so LoS probability – much sooner than a 20 km-altitude HAPS does. The usable-relay radius, where access-hop capacity crosses a 0.5 bits/s/Hz floor, is ≈ 6.5 km; averaged over azimuth around the serving HAPS, relay capacity does not exceed the direct link's at any distance tested (0–100 km) once averaged this way. The relay therefore helps only the small fraction of users within this radius of the one fixed UAV, not every user – a single static platform cannot be close to everyone across a 100 km-radius footprint; §X quantifies how many additional, well-placed UAVs this implies. This motivates including the UAV relay as an association option for the learned policy of §VIII, which (§IX) learns to invoke it only for users it actually helps.

VI. SIMULATION RESULTS

All results use a Python simulation stack (`NumPy/pandas/Matplot`
atmospheric and scintillation losses use the ITU-Rpy (`itur`) [24] library's implementation of the relevant ITU-R recommendations.

A. WHY 38 GHZ: FREQUENCY SELECTION

This subsection validates the 38 GHz backhaul/feeder choice already fixed in §III (WRC-19 allocation). It is included to address a natural reader question: why not

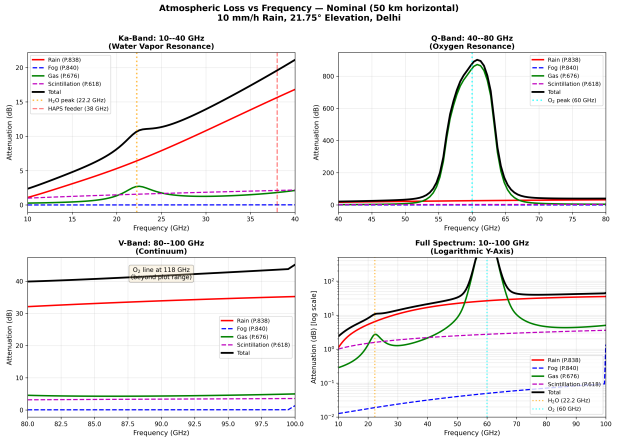


FIGURE 2: Atmospheric loss (rain, fog, gas) vs. frequency across the Ka/Q/V bands for the Delhi gateway path at 10 mm/h rain.

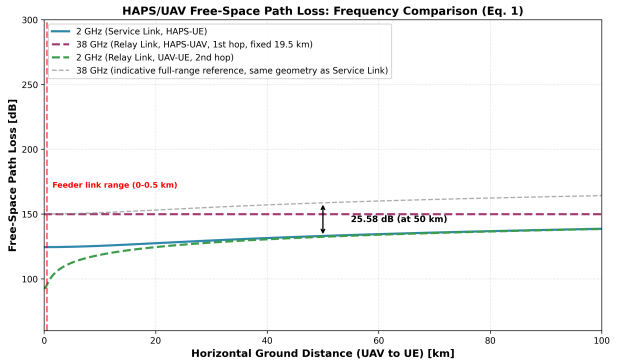


FIGURE 3: FSPL vs. distance (Eq. (1)) – same functional form for 2 and 38 GHz; the 38 GHz feeder sits ≈25.6 dB higher, purely from the $20\log_{10}(38/2)$ frequency term.

simply pick a lower-loss frequency instead of designing around rain, fog, and gas attenuation at 38 GHz?

Rain loss (ITU-R P.838/P.618 [4], [5]) is the dominant term and grows roughly linearly with frequency, up to ≈19 dB at 99.9 % availability. Fog/cloud (ITU-R P.840 [7]) is negligible below 10 GHz. Gas absorption (ITU-R P.676 [6]) has two resonance peaks to avoid: water vapor at 22.2 GHz and oxygen at 60 GHz, where total loss exceeds 800 dB over 50 km and the band is unusable. **38 GHz sits in the valley between these peaks** (≈0.2–0.3 dB/km clear-sky gas loss) – this, not just available bandwidth, is why it was chosen over a frequency closer to either resonance.

B. PATH LOSS AND SINR VS. DISTANCE

C. ANGULAR INTERFERENCE SECTORS AND SCINTILLATION

The good/bad-sector result of §V (Fig. 6) is one half of the angle- and weather-dependent picture; scintillation (Eq. (3)) is the other. At 30° elevation, $p = 0.01\,\%$, scintillation fade is ≈0.49 dB at 38 GHz versus ≈0.09 dB at 2.1 GHz – growing with frequency and shrinking with elevation. Both effects favor positioning the UAV relay near nadir (higher elevation) when feasible, especially on the HAPS–UAV feeder hop, which also tends to fall in the interferer-facing sector's complement. This is not, however, a single clean optimization: each HAPS can simultaneously be a serving node (near-nadir, low-loss, for its own users) and an interferer for the other two HAPS' users – jointly optimizing positioning across both roles is left to future work (§X).

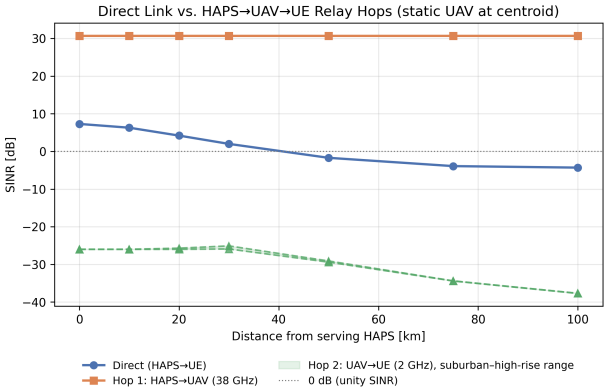


FIGURE 4: SINR vs. distance from the serving HAPS, azimuth-averaged, for the direct link and both relay hops, with the UAV fixed at the HAPS triangle's centroid. The feeder hop (HAPS→UAV, 38 GHz) is flat, since the UAV's distance from each HAPS is fixed by the constellation's symmetry; the access hop (UAV→UE, 2 GHz) instead falls with each user's real, azimuth-dependent distance from that one point, collapsing within a few km (§V-A).

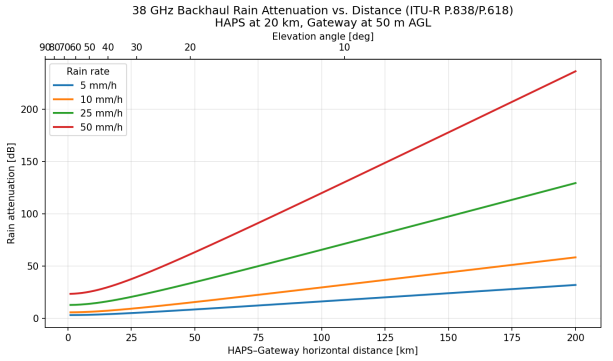


FIGURE 5: 38 GHz backhaul rain attenuation vs. HAPS–Gateway horizontal distance (ITU-R P.838/P.618), grows faster than linear past ≈100 km as elevation drops and the slant path lengthens (§IV-A).

D. WHY LEARNING IS NEEDED: SPATIAL SINR HETEROGENEITY

At a fixed 50 km radius, SINR varies by up to 9.94 dB with angular position (§V), while doubling the link

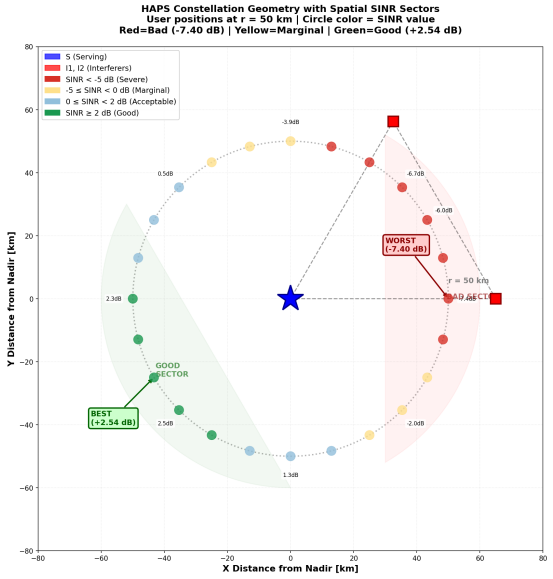


FIGURE 6: HAPS constellation geometry (65 km equilateral triangle: S at origin, I1 at 0°, I2 at 60°) with spatial SINR sectors at $r = 50$ km. Good sector (green, ∼120–290°) vs. bad sector (red, ∼330–75°). Best +2.54 dB at 210°; worst −7.40 dB at 0°.

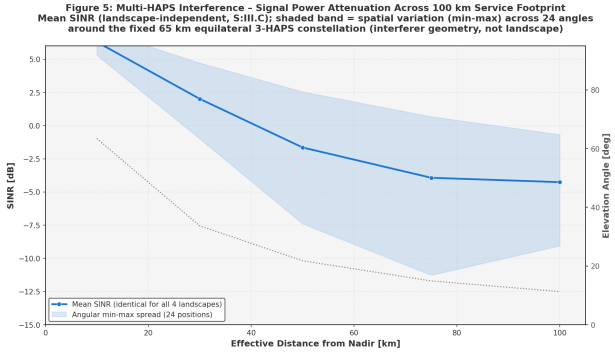


FIGURE 7: Mean SINR and angular min–max spread across the 100 km service footprint: falls from +6.3 dB at 10 km to −4.3 dB at 100 km, while the spread widens to 11.94 dB at 75 km.

distance costs only ∼2–3 dB – angular position affects SINR more than distance does. A simple "always pick max-SINR" rule cannot anticipate these angular patterns or adapt as weather and battery state change, motivating a learned, real-time association policy.

E. STAGE 2: BASELINE ASSOCIATION POLICIES

Before introducing a learned policy, three rule-based (non-learned) association policies are evaluated on the channel model above, to establish how much association choice alone – without any learning – already changes outcomes. Each of N users chooses one of six options every hour: a direct link to HAPS1–3, or a UAV relay via HAPS1–3 (§V-A).

TABLE 2: LSTM (early-stopped at the epoch validated on held-out 2025, §IX) against persistence on 2026 (6556 forecast hours, no 2026 data in training): MAE/RMSE in dB per horizon. Bold marks the better method.

Horizon	LSTM MAE	Persist. MAE	LSTM RMSE	Persist. RMSE
+1 h	0.15	**0.12**	0.54	**0.53**
+2 h	0.21	**0.20**	**0.70**	0.75
+3 h	**0.24**	0.26	**0.80**	0.90
+4 h	**0.27**	0.30	**0.86**	1.02
+5 h	**0.28**	0.34	**0.90**	1.11
+6 h	**0.29**	0.36	**0.92**	1.17

- **Random**: uniform-random choice among the six options each hour (sanity floor).
- **Max-SINR**: always the best-SINR *direct* link (Eq. (7)); cannot use the relay, no memory of battery or fairness state.
- **Max-throughput**: the highest-throughput option among all six every hour, assuming perfect, instantaneous knowledge of every option's SINR and capacity – a theoretical upper bound, not a deployable policy.

These three policies are carried forward as the baselines in §IX, where they are compared against the learned LSTM+DQN policy of §VIII on the same held-out days and the same six-action set.

VII. LSTM RAIN-ATTENUATION FORECASTER

The DQN must choose an association at hour t before that hour's rain is observed, so it needs a forecast of the feeder loss. An LSTM is trained to map the past 24 hours of weather (14 features: temperature, humidity, pressure, wind, precipitation, cloud amount, visibility, rain/fog codes, cloud liquid water, and sine/cosine encodings of time-of-day and season, from hourly NASA POWER, METAR and ERA5 data for Delhi, 2023–2026) to the next six hourly values of HAPS-to-gateway rain attenuation (38 GHz, 89.86° elevation, ITU-R P.838/P.618) – the only link in this model where rain is a significant, time-varying impairment (§IV-B); the 2 GHz service/relay link's rain, fog, and gas loss are negligible, so no forecast is needed there. Forecasts are always evaluated out-of-sample (e.g. 2026 forecasts from a model trained only on 2023–2025), and a forecast for hour t is issued at $t-1$, so the agent never sees the rain of the hour it is acting on. In deployment this would be a continuous, incremental feed, with the model updating as new weather data arrives.

An earlier version of this model was trained for a fixed 30 epochs; validation on held-out 2025 data (never touching 2026 until the final evaluation above) showed training loss falling monotonically throughout while validation RMSE was best around epoch 4 and only got worse afterward – a textbook overfitting signature, not a sign the task itself is unlearnable. Early-stopping at the validated epoch turns a mostly-losing comparison into a mostly-winning one: persistence is now more accurate only at +1 h (and narrowly, on MAE, at +2 h); the LSTM

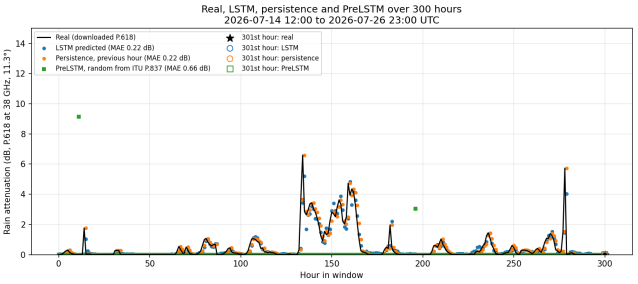


FIGURE 8: Real, LSTM, persistence and a synthetic PreLSTM baseline over a 300-hour window in July 2026 (the window with the most rain). LSTM: one-hour-ahead forecast. Persistence: previous hour's value. PreLSTM: random hourly rain rates drawn from ITU-R P.837 statistics, converted via P.618.

is more accurate at every other horizon and measure (Table 2). The DQN's state uses all six forecast horizons regardless, since the gain is not uniform across horizons but the longer horizons matter most for pre-emptive rerouting. **Observation:** with this improved model, a per-horizon-weighted blend of LSTM and persistence (weights chosen on 2025 validation, never on the 2026 test set) adds a small further improvement only at +1 h (0.526 vs. 0.531 persistence-only and 0.566 LSTM-only dB RMSE); at every other horizon the validated weight is ≥ 0.99 on the LSTM, i.e. persistence no longer has anything to add once the LSTM itself is properly trained.

Whether this forecast accuracy actually translates into a better learned association policy is a separate question, addressed once the DQN itself has been defined (§IX).

VIII. DQN ASSOCIATION AGENT

A. PER-USER MDP FORMULATION

Each of the three platforms can serve a user directly or through the UAV relay, giving six actions $a_k \in \{0, \ldots, 5\}$ (actions 0–2: direct HAPS1–3; actions 3–5: UAV relay via HAPS1–3). With N users, a centralized MDP over joint assignments is intractable, so the problem decomposes into per-user MDPs: each user independently chooses a platform, and reward is computed after all assignments.

With a single, static relay UAV (Table 1), choosing *which* HAPS feeds the UAV's feeder hop is folded into this same per-user action set, effectively treating the UAV as if it were just another user requesting a serving HAPS. Only the *shape* of that choice is shared (pick one of three HAPS) – the channel model evaluating it is not: the UAV's feeder hop still uses the 38 GHz, ITU-R P.618 feeder model of §IV-A (its own antenna gains and atmospheric terms), never the 2 GHz building-blockage model of §IV-B that a direct user's choice is evaluated against. Sharing the action's shape, not its physics, is a fair simplification when there is only one UAV to assign. This does not generalise to a UAV fleet: at fleet scale, UAV-to-HAPS assignment (an O(fleet size) problem, e.g. the ∼200-UAV deployment of §X) and UE-to-UAV/HAPS access (an $O(N)$ problem) become two separably-sized assignment problems that cannot stay bundled into one per-user action, since the UAV side would then be a shared resource contended for by many users' decisions at once rather than belonging to any single user's action space.

State (per user, 21-dimensional): SINR to each platform (dB, Eq. (7)), horizontal distance to each (km), each HAPS's battery state of charge (%, §VIII-B), the UAV-relay capacity to each platform (bits/s/Hz, §V-A), rainfall rate, visibility, time-of-day, and the six LSTM rain forecasts $\hat{A}_t, \ldots, \hat{A}_{t+5}$ (§VII), issued at $t-1$ so no rain of hour t enters the state. Tropospheric scintillation (§IV-A) is not included as a state feature – it is a smaller, time-varying effect ($\leq$0.6 dB at 38 GHz, $p = 1\,\%$) whose forecasting is left to future work.

Reward. For user k with linear SINR $\gamma_k = 10^{\mathrm{SINR}_{k,\mathrm{dB}}/10}$ on its chosen link,

$$r_k = \omega_1 T_k c_{\mathrm{rain}} + P_k^{\mathrm{sev}} + P_k^{\mathrm{deg}} + \omega_2 J_{\mathrm{fair}} - \omega_3 P_{\mathrm{eng}} + P_{\mathrm{LB}},$$
$$T_k = \log_2(1+\gamma_k), \qquad c_{\mathrm{rain}} = 10^{-A_{\mathrm{rain}}/10} \in [0.05, 1],$$
$$J_{\mathrm{fair}} = \frac{(\sum_k \gamma_k)^2}{N \sum_k \gamma_k^2}, \qquad P_{\mathrm{eng}} = \frac{1}{3}\sum_{p=1}^{3}\left(1 - \frac{\mathrm{SoC}_p}{100}\right), \tag{9}$$

with $\omega_1 = 0.5, \omega_2 = 0.3, \omega_3 = 0.2$; c_{rain} de-rates every user's throughput credit alike, whether served directly or via the UAV relay, because all traffic – direct or relayed – ultimately transits the rain-faded 38 GHz backhaul to reach the core network (§III). c_{rain} leaves the other reward terms (fairness, outage, energy) untouched. Penalties (per user): severe outage $P_k^{\mathrm{sev}} = -100$ if $\mathrm{SINR}_{k,\mathrm{dB}} < -2$ dB; degraded service $P_k^{\mathrm{deg}} = -20$ if $-2 \leq \mathrm{SINR}_{k,\mathrm{dB}} < 0$ dB; and a shared low-battery penalty $P_{\mathrm{LB}} = -50$ for any platform with $\mathrm{SoC}_p < 10\,\%$ and at least one assigned user. The reported fairness metric (Jain index [28] over per-user delivered throughput, $T_k c_{\mathrm{rain}}$) differs from J_{fair} above, which is Jain over linear SINR and enters only the reward.

B. BATTERY MODEL

Each HAPS's battery state of charge (SoC) is tracked hourly,

$$\Delta\mathrm{SoC} = \frac{P_{\mathrm{solar}}(t) - (P_{\mathrm{backhaul}} + P_{\mathrm{payload}} + \sum_k P_{k,\mathrm{user}})}{C_{\mathrm{battery}}}, \tag{10}$$

with $P_{\mathrm{backhaul}} = 500$ W, $P_{\mathrm{payload}} = 200$ W, and $C_{\mathrm{battery}} = 12\,000$ Wh. Because the HAPS operates above >99 % of clouds, solar generation is independent of ground weather and follows a diurnal cosine profile,

$$I_0(t_{\mathrm{hr}}) = \begin{cases} 600\cos(\pi(t_{\mathrm{hr}} - 12)/12) & 6 \leq t_{\mathrm{hr}} \leq 18 \\ 0 & \text{night} \end{cases} \ [\mathrm{W/m^2}], \tag{11}$$

giving $P_{\mathrm{solar}}(t) = 20\,\mathrm{m}^2 \times 0.20 \times I_0(t)$, i.e. 2400 W at noon and 0 W at night. SoC is clipped to $[0, 100]$ %;

TABLE 3: Policy comparison, 50 held-out 2025 days, mean over ten experiments (std. dev. across experiments in parentheses). Jain is over per-user delivered throughput.

Policy	Reward/step	Jain	Outage/50
Random (six actions)	−91.9 (1.5)	0.262 (0.005)	41.5 (0.3)
Max-SINR (direct only)	−13.8 (1.7)	**0.865** (0.002)	0.8 (0.2)
Max-throughput (six actions)	**−13.5** (1.6)	0.862 (0.002)	**0.7** (0.2)
LSTM + DQN	−14.6 (1.8)	0.857 (0.005)	1.3 (0.4)

below 5 % the platform shuts down (no service), and below 10 % with assigned users the reward's low-battery penalty applies, so the agent must learn to balance load across platforms to avoid over-draining any one HAPS overnight.

IX. DQN RESULTS

The three stage-2 baseline policies (§VI-E: random, max-SINR, max-throughput) are now compared against the stage-3 LSTM+DQN agent (§VIII) on the same held-out days and six-action set. The DQN is a custom PyTorch (`torch/torch.nn`) network, trained on 2400 rolling 24-hour episodes (cycling the 400 real 2024 episodes six times; 400 was the original budget, but with the UAV relay fix above making three of the six actions traps for most users, training needed longer to converge – 400 episodes left reward still improving noisily) and evaluated on 50 held-out 2025 days, across ten experiments (random seeds).

With the UAV relay correctly modeled as a single static platform (§V-A), the picture changes from an earlier version of this model that let the UAV hover beside whichever user it served: the random policy now suffers badly, since three of its six equally-likely actions are usually traps, and **LSTM+DQN no longer has the highest reward** (Table 3) – it is statistically indistinguishable from, and on raw reward slightly behind, both max-SINR and max-throughput (paired across the same ten seeds: −0.73 vs. max-SINR, $p < 0.0001$, 0/10 seeds favoring the DQN; −1.07 vs. max-throughput, $p < 0.0001$, 0/10). Decomposing the reward by term shows the gap is driven almost entirely by a higher severe-outage rate: the DQN leaves users in severe outage (SINR < −2 dB) on 2.5 % of user-hours, against 1.3–1.7 % for the two rule-based policies, which by construction never choose worse than the best currently-visible direct option. The throughput and fairness terms do differ too, at conventional significance ($p < 0.05$), but by amounts an order of magnitude smaller than the severe-outage term, and battery energy is identical across all three policies (the energy model does not depend on association policy); the severe-outage term, net of a smaller advantage the DQN holds on *degraded*-outage rate, accounts for nearly all of the reward gap. Max-SINR needs only a real-time

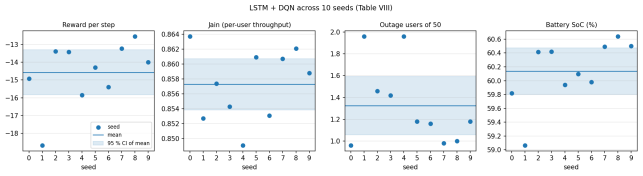


FIGURE 9: LSTM+DQN, ten independent seeds (dot = one training run; line = mean; band = 95 % CI of the mean). Reward/step -14.57 ± 1.78 (95 % CI $[-15.84, -13.30]$); Jain 0.857 ± 0.005; outage 1.33 ± 0.37 of 50 users; battery SoC 60.14 ± 0.47 %. The shaded band narrows with more seeds and reflects precision of the estimated mean, not run-to-run spread; the tight spread across all ten seeds shows the policy is reliably learned to this level, not a lucky or unlucky single run – consistency, not the reward level itself, is what this figure demonstrates.

SINR measurement, already a deployable rule; max-throughput, which additionally compares all six options' instantaneous capacity including relay, needs perfect, instantaneous knowledge of capacities it cannot actually measure non-causally, but since relay rarely wins post-fix, it also collapses in practice to "pick the best direct option." Despite this, the DQN is the only policy that retains and actually uses the relay path: it invokes relay specifically for users whose best direct SINR already averages −1.9 dB (population mean +2.4 dB), consistent across all ten seeds ($p = 2.4 \times 10^{-13}$) – a targeted, structural use of relay capacity that neither rule-based baseline can replicate, since Max-SINR cannot use relay at all and Max-throughput's instantaneous rule has no notion of *which* users most need it.

Does the rain forecast help the DQN itself? A forecast ablation, trained and evaluated with this same ten-seed, 2400-episode protocol (`lstm` vs. a `none` variant with the six rain-forecast features zeroed, and an `oracle` variant given the true future attenuation) finds `lstm`, `none`, and `oracle` statistically indistinguishable from each other on reward, fairness, and outage (all $p > 0.1$). With the now properly early-stopped forecast (§VII), the earlier, less accurate forecast's small negative effect on reward is gone – but no positive effect has taken its place, even against the `oracle` ceiling. This is not evidence against forecasting in general, nor even against this rain forecast's own accuracy, which is independently verified in §VII; it reflects that this particular action space gives the agent no rain-reactive lever to exploit. Rain already shows up in the state as its effect on current SINR, and both links the agent can act on are largely insulated from rain in the first place: the 2 GHz service and access hops are physically resilient to rain at that frequency, and the 38 GHz feeder hop's ≈19 dB of rain attenuation (§IV-A) is already absorbed by the link's ≈18.6 dB of design margin, which comes from the modeled hardware

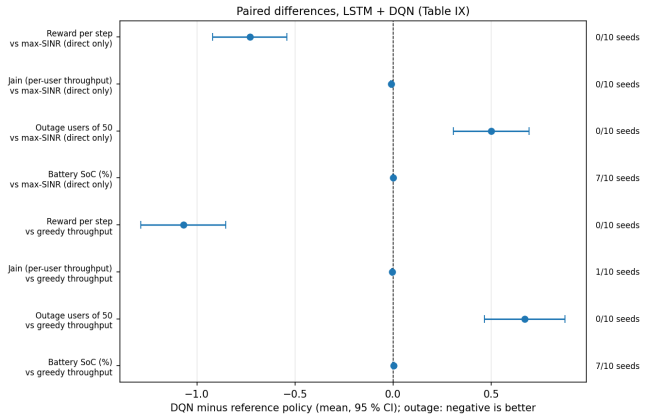


FIGURE 10: Paired differences, LSTM+DQN minus baseline, per experiment (50 days each, ten seeds); whiskers crossing zero are not significant. Vs. max-SINR: reward -0.73 ($p < 0.0001$, 0/10 seeds favoring DQN), Jain -0.0079 ($p = 0.0003$, 0/10), outage $+0.50$, i.e. worse ($p = 0.0002$, 0/10), battery SoC ≈ 0 ($p = 0.19$, tie). Vs. max-throughput: reward -1.07 ($p < 0.0001$, 0/10), Jain -0.0043 ($p = 0.0099$, 1/10), outage $+0.67$, i.e. worse ($p < 0.0001$, 0/10), battery SoC ≈ 0 ($p = 0.09$, tie). The DQN now loses to both rule-based policies on reward, fairness, and outage on every seed (or all but one), and ties them only on battery management; §IX decomposes this gap and the relay-targeting behaviour that nonetheless distinguishes the DQN from either baseline.

parameters (Tx power, antenna gain) rather than from any choice made in this paper. A forecast could only earn its keep here if some action let the agent act *ahead of* a link's degradation rather than react to it once visible – a capability this fixed-geometry, six-action MDP does not have.

X. CONCLUSION

This paper characterised the HAPS–UAV–Gateway air-to-ground channel across its three link types, quantified multi-HAPS interference and the UAV-relay path as an alternative to the direct service link, and used that channel model to train and evaluate a DQN association agent.

Channel model. Spatial SINR heterogeneity – up to 9.94 dB at a fixed 50 km radius, driven by angular position relative to the two interfering HAPS – dominates service-link quality; the backhaul link remains noise-limited and largely unaffected by constellation geometry.

Relay path. With the single relay UAV correctly modeled as static (§V-A), fixed at the HAPS triangle's centroid, its feeder hop is reliably flat (10.2 bits/s/Hz, ≈27.5° elevation, the same for every HAPS by the constellation's symmetry) but its access hop collapses within a few kilometres of that one fixed point, far faster than the direct link degrades with distance from a HAPS. Averaged over azimuth, relay capacity does not exceed the direct link's at any distance tested, so a single static UAV helps only the small minority of users within its ≈6.5 km usable radius – a covering-number estimate puts full, gap-free coverage of the 100 km service radius at ∼200–270 such UAVs (§VIII discusses why that is a separate fleet-assignment problem, not a scaled-up version of today's single-UAV model). The direct service link itself does not scale to replace this: it has no modeled mechanism for sharing its narrow band across many simultaneous users, unlike the feeder link, which is inherently wideband and aggregates traffic the way terrestrial backhaul does. Both links are, in that sense, best read as a wide-area, low-infrastructure fallback (direct) and a capacity-scalable but currently under-deployed primary path (relay), rather than the direct link being primary and relay the backup, as a single static UAV's limited reach might otherwise suggest.

Learned association. With the relay path corrected, the LSTM-fed DQN is statistically indistinguishable from, and on raw reward and outage slightly behind, both max-SINR and max-throughput – a reversal of an earlier result obtained under the unrealistic always-nearby-UAV assumption. The gap traces almost entirely to a higher severe-outage rate, not to battery management, on which all three are identical by construction. The rain forecast itself shows no measurable reward benefit to the trained policy, not even against an oracle given the true future attenuation (§IX) – but this is a statement about this action space, not about rain forecasting in general or this forecast's own accuracy, which is independently verified in §VII. Both links available to the agent are already largely insulated from rain: the 2 GHz hops are physically resilient to it, and the 38 GHz feeder's rain attenuation is already absorbed by the link's design margin (itself a consequence of the modeled hardware parameters, not a choice made in this work); a forecast only pays off where an action lets the agent move *ahead of* degradation, which a fixed-geometry, six-action MDP does not offer. What distinguishes the DQN is structural, not a reward-table win: it is the only policy that retains and actually uses the relay path, and it does so as a targeted fallback for users already near the outage threshold, not indiscriminately – a capability neither rule-based baseline can have, since Max-SINR cannot use relay at all and Max-throughput's instantaneous rule has no notion of which specific users need it most.

Roadmap for future investigations.

1) Model multiple UAVs.

- *HAPS selection for UAVs.* This paper already shows relaying is worthwhile mainly within ≈6.5 km of its single UAV, and the DQN learns to target it there (§IX). Once a fleet of non-equidistant UAVs replaces that one fixed platform, *which* HAPS feeds each UAV becomes a real choice – a UAV-to-HAPS assignment problem (O(fleet size)) that needs its

own MDP, distinct from the $O(N)$ UE-to-access-point assignment already in §VIII, with relay-vs-direct decided as its own explicit policy rather than folded into one per-user choice.

- *Relative motion of UAVs and HAPS.* Model movement of both platforms, and relative to each other and to users, rather than the static geometry assumed throughout this paper.

2) LSTM improvements.

- *Alternative architectures.* A ConvLSTM on gridded reanalysis data, a GRU as a lighter-weight alternative, or a spatio-temporal graph model such as DCRNN, T-GCN or Graph WaveNet if future sites are irregularly placed.
- *Ensembling.* A validated, per-horizon-weighted blend of the LSTM and persistence forecasts (§VII) adds a small further improvement at the shortest horizon once the LSTM itself is properly early-stopped; worth revisiting if future sites or seasons shift which horizons persistence remains competitive at.
- *Forecasting a different impairment.* The DQN ablation (§IX) found no action in this fixed-geometry MDP that benefits from a rain forecast specifically. Multi-year, hourly *time-series* data was only available for rain rate here; fog liquid water content, scintillation structure constants, and gas absorption are instead computed instantaneously from climatology in this work (ITU-R P.840/P.618/P.676), not forecast. Whether forecasting one of those pays off would depend on both data availability and the agent having an action that can actually act ahead of that specific impairment – neither of which held for rain in this MDP.

3) Dynamic topology and geometry.

- *Changing HAPS roles.* Shift a node between serving and interfering based on its distance to the gateway, and jointly optimize UAV/HAPS positioning across both roles a node could play at once, rather than each in isolation.
- *UAV positioning to avoid outages and losses.* Choose UAV placement to avoid the blind-corner coverage gaps a naive non-overlapping tiling leaves behind, reducing the resulting edge-of-coverage errors.
- *Beamforming.* Consider beamforming between HAPS and gateway as geometry changes.

XI. ABBREVIATIONS

ABS	Aerial Base Station
AGL	Above Ground Level
CDF	Cumulative Distribution Function
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
FDMA	Frequency Division Multiple Access
FSPL	Free-Space Path Loss
GW	Gateway
HAPS	High-Altitude Platform Station
ITU	International Telecommunication Union
K-band	18–27 GHz (ITU-R P.618 scintillation model)
Ka-band	27–40 GHz (Scaled from K-band via $(f/20)^{0.7}$)
LoS	Line-of-Sight
LSTM	Long Short-Term Memory
LWC	Liquid Water Content
MDP	Markov Decision Process
MLP	Multi-Layer Perceptron
MVP	Minimum Viable Product
NLoS	Non-Line-of-Sight
PDF	Probability Density Function
PL	Path Loss
RH	Relative Humidity
SINR	Signal-to-Interference-plus-Noise Ratio
SoC	State of Charge (Battery)
TDMA	Time Division Multiple Access
TD	Temporal Difference
UAV	Unmanned Aerial Vehicle
UTD	Uniform Theory of Diffraction

XII. REFERENCES

REFERENCES

[1] ITU-R Recommendation M.2101-0, "Characteristics of High Altitude Platform Stations (HAPS) operating in the stratosphere," 2017. Available: https://www.itu.int/rec/R-REC-M.2101/latest/
[2] ITU-R Recommendation M.2106-1, "Characteristics of High Altitude Platform Stations (HAPS) operating in the stratosphere," 2012. Available: https://www.itu.int/rec/R-REC-M.2106/latest/
[3] ITU-R Recommendation P.530-17, "Propagation data and prediction methods required for the design of terrestrial line-of-sight systems," 2021. Available: https://www.itu.int/rec/R-REC-P.530/latest/
[4] ITU-R Recommendation P.618-13, "Propagation data and prediction methods for the design of Earth-space telecommunication systems," 2021. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.618-13-202108-S!!PDF-E.pdf [Covers rain attenuation reduction factors, slant-path geometry, scintillation model]
[5] ITU-R Recommendation P.838-3, "Specific attenuation model for rain for use in prediction methods," 2005. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.838-3-200503-S!!PDF-E.pdf [Annex 1: frequency-dependent rain coefficients $k(f)$, $\alpha(f)$]
[6] ITU-R Recommendation P.676-12, "Attenuation by atmospheric gases and related effects," 2021. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.676-12-202112-S!!PDF-E.pdf [Section 3.1.2: O_2 resonance (59.8 GHz, 118.75 GHz); Section 3.2.1: H_2O resonance (22.235 GHz)]
[7] ITU-R Recommendation P.840-6, "Attenuation due to clouds and fog," 2021. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.840-6-202101-I!!PDF-E.pdf [Liquid water content model for fog/cloud attenuation]

[8] ITU-R Recommendation P.835-7, "Reference standard atmospheres," 2017. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.835-7-201703-I!!PDF-E.pdf

[9] ITU-R Recommendation P.836-5, "Surface water vapour statistics," 2017. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.836-5-201703-I!!PDF-E.pdf

[10] ITU-R Recommendation P.837-7, "Characteristics of precipitation for propagation modelling," 2017. Available: https://www.itu.int/dms_pubrec/itu-r/rec/p/R-REC-P.837-7-201706-I!!PDF-E.pdf

[11] ITU-R Recommendation P.1410-5, "Propagation data and prediction methods for the design of terrestrial broadband radioaccess systems operating in a frequency range from 3 to 60 GHz," 2021. Available: https://www.itu.int/rec/R-REC-P.1410/latest/ [Elevation-angle-dependent LoS probability baseline]

[12] 3GPP TR 38.821, "Study on New Radio (NR) and its Evolution for Non-Terrestrial Networks (NTN)," Release 17, September 2022 (study report; background only). Available: https://www.3gpp.org/DynaReport/38821.htm

[13] 3GPP TS 38.214, "NR; Physical layer procedures for data," Release 16, 2021. Available: https://www.3gpp.org/DynaReport/38214.htm [CQI/MCS-to-SINR design reference: CQI index 1 (QPSK, code rate 78/1024) requires SINR ≈ -6 dB; CQI index 7 (16-QAM, code rate 378/1024) requires SINR $\approx +5$ dB; CQI index 15 (256-QAM, code rate 948/1024) requires SINR $> +22$ dB, all at the standard 10% first- transmission BLER target.]

[14] B. Karaman, I. Basturk, F. Kara, E. Zeydan, E. A. Beyazit, S. Taskin, E. Björnson, and H. Yanikomeroglu, "On-Demand HAPS-Assisted Communication System for Public Safety in Emergency and Disaster Response," IEEE Commun. Mag., 2025. arXiv:2507.09153. [Ka-band (30 GHz) HAPS backhaul link budget, Table I: Tx power 43.2 dBm, Tx antenna gain 13.8 dBi, Rx (VSAT) antenna gain 39.7 dBi – used here as the literature-typical antenna-gain basis for this document's 38 GHz feeder-style links, §V-A.]

[15] K. Anim, J.-N. Lee, and Y.-B. Jung, "High-Gain Millimeter-Wave Patch Array Antenna for Unmanned Aerial Vehicle Application," Sensors, vol. 21, no. 11, p. 3914, 2021. DOI: 10.3390/s21113914 [UAV-mountable 32×32 mmWave patch array antenna, realized gain 30.7–32.8 dBi at 25.4–26.9 GHz – basis for this document's UAV-side receive antenna gain on the HAPS $\rightarrow$ UAV hop, §V-A.]

[16] 3GPP TS 36.104, "Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) Radio Transmission and Reception," Release 16, December 2021. Available: https://www.3gpp.org/DynaReport/36104.htm [LTE channel bandwidth specifications; 20 MHz service-link allocation]

[17] A. H. Arani, P. Hu, and Y. Zhu, "HAPS-UAV-Enabled Heterogeneous Networks: A Deep Reinforcement Learning Approach," IEEE Open J. Commun. Soc., vol. 4, pp. 1745–1760, 2023. DOI: 10.1109/OJCOMS.2023.3294598

[18] J. Holis and P. Pechac, "Elevation Dependent Shadowing Model for Mobile Communications via High Altitude Platforms in Built-up Areas," IEEE Trans. Antennas Propag., vol. 56, no. 4, pp. 1078–1084, Apr. 2008. DOI: 10.1109/TAP.2008.919209 [Building-blockage LoS probability series; environment parameters (Table I): (α, β, ξ) for suburban/urban/dense-urban/high-rise]

[19] A. Bakambekova, N. Kouzayha, and T. Al-Naffouri, "On the Interplay of Artificial Intelligence and Space-Air-Ground Integrated Networks: A Survey," IEEE Open J. Commun. Soc., vol. 5, pp. 4613–4673, 2024.

[20] European Communications Committee (ECC), "Minimum performance characteristics for spectrum sharing between Fixed Satellite Service (FSS) and other services," ECC Report 156, April 2011 (updated December 2014). Available: https://www.ecc.cept.org/ecc/documents/ecc-reports/ [Spectrum-sharing guidance; rain attenuation margins at Ka-band]

[21] European Communications Committee (ECC), "Radio Frequency Channel Arrangements for Fixed Service Systems Operating in the Bands 6–7 GHz and 36–40 GHz," ECC Recommendation (04)05, Strasbourg, France. [Ka-band channelization for backhaul systems]

[22] A. Goldsmith, Wireless Communications, Cambridge University Press, 2005. [Multi-user SINR in linear domain; noise and interference are additive in linear scale, not dB]

[23] 3GPP, "Study on New Radio (NR) and its evolution for Non-Terrestrial Networks," Technical Report 3GPP TR 38.821, Release 16, April 2020. [Multi-beam HAPS interference model: $\mathrm{SINR}_k = P_{\mathrm{serving}} \cdot G_k / (N_0 + \sum P_i \cdot G_{i,k})$]

[24] I. Portillo-Ramirez, "itur: Python implementation of ITU-R P recommendations," GitHub, 2024. Available: https://github.com/iportillo/itur [Bundled ITU-R models for rain, fog, gas, scintillation]

[25] V. Mnih, K. Kavukcuoglu D. Silver, A. A. Rusu, J. Veness, G. Bellemare, A. Graves, M. Riedmüller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," Nature, vol. 529, pp. 529–533, 2015. [Foundational DQN algorithm: experience replay, target networks, Q-learning]

[26] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA: MIT Press, 2018. [MDP theory, value functions, policy gradient methods; Chapter 6: Q-learning]

[27] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv preprint arXiv:1707.06347, 2017. [Policy-gradient alternative to DQN; higher sample complexity for discrete actions]

[28] R. K. Jain et al., "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," *arXiv preprint arXiv:cs/9809099*, 1984.

[29] ITU, Final Acts of the World Radiocommunication Conference (WRC-19), agenda item 1.4 (HAPS): identification of 31–31.3 GHz and 38–39.5 GHz for HAPS use, and of 47.2–47.5 GHz and 47.9–48.2 GHz. [Verify resolution number and footnote wording against the Radio Regulations.]

[30] 3GPP TS 38.101-5, "NR; User Equipment (UE) radio transmission and reception; Part 5: Satellite access Radio Frequency (RF) and performance requirements." Band n256 (FDD): 1980–2010 MHz uplink, 2170–2200 MHz downlink. [Verify release number.]

[31] G. Svistunov, A. Akhtarshenas, D. López-Pérez, M. Giordani, G. Geraci, and H. Yanikomeroglu, "Bridging Earth and Space: A Survey on HAPS for Non-Terrestrial Networks," IEEE Commun. Surveys Tuts., under review, 2025. arXiv:2510.19731.

[32] T. Naous, M. Itani, M. Awad, and S. Sharafeddine, "Reinforcement Learning in the Sky: A Survey on Enabling Intelligence in NTN-Based Communications," IEEE Access, vol. 11, pp. 19941–19968, 2023. DOI: 10.1109/ACCESS.2023.3249890

[33] S. Liu, H. Dahrouj, A. Kammoun, and M.-S. Alouini, "Sum Rate and Worst-Case SINR Optimization in Multi-HAPS–Ground Integrated Networks," IEEE Trans. Aerosp. Electron. Syst., vol. 62, pp. 4540–4555, 2026.

[34] A. Cornejo, S. Landeros, J. M. Matías, F. G. Ortiz-Gomez, R. Martínez, and M. A. Salas-Natera, "Method of Rain Attenuation Prediction Based on Long–Short Term Memory Network," Neural Process. Lett., vol. 54, no. 4, pp. 2959–2995, 2022. DOI: 10.1007/s11063-022-10749-1